BigData Spark Hands-On 5 [part2]

launching spark on k8s

arnaud.nauwynck@gmail.com

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Outline

part 1/2

- Install podman (~Docker) on local Windows/Mac/Linux
- Install Kubernetes Cluster "Kind" (=K8s in Docker)
- Install Kubernetes Tools: k9s, helm, aliases
- Install spark-operator for K8s using Helm + check/fixes

(Optional) Launch spark-shell on Docker, in mode "--master local[*]"
(Optional) Launch spark with "spark-submit --master k8s://localhost:port"

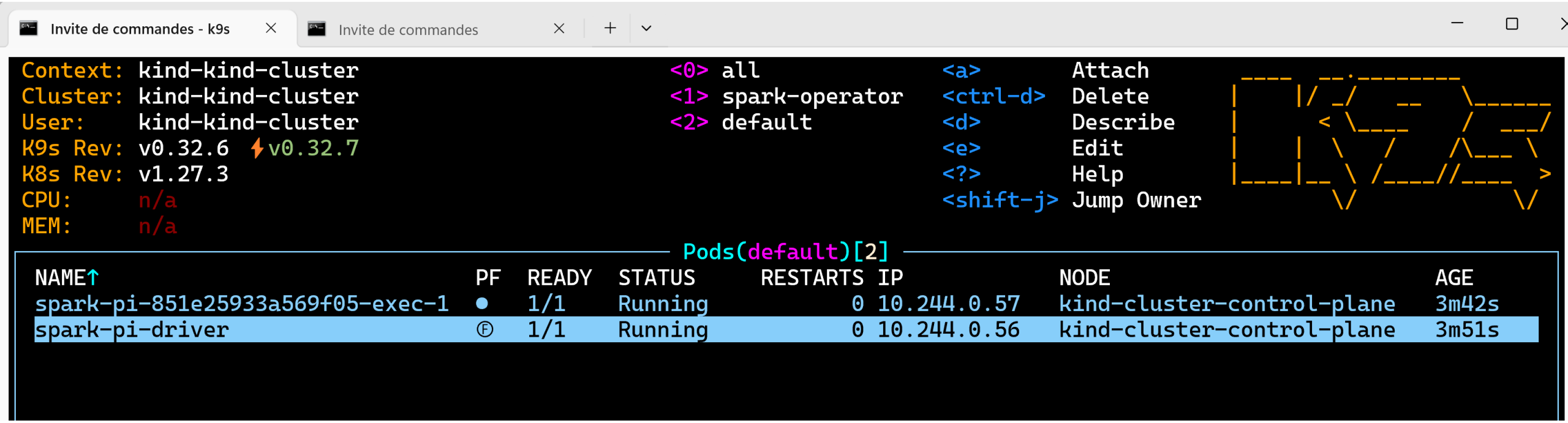
Launch spark with "kubectl apply -f file.yaml"

(Optional... and backup if local install fails) Using Azure Kubernetes

"KISS" = Keep It Stupid Simple

kubectl apply -f spark-pi.yaml

Check running

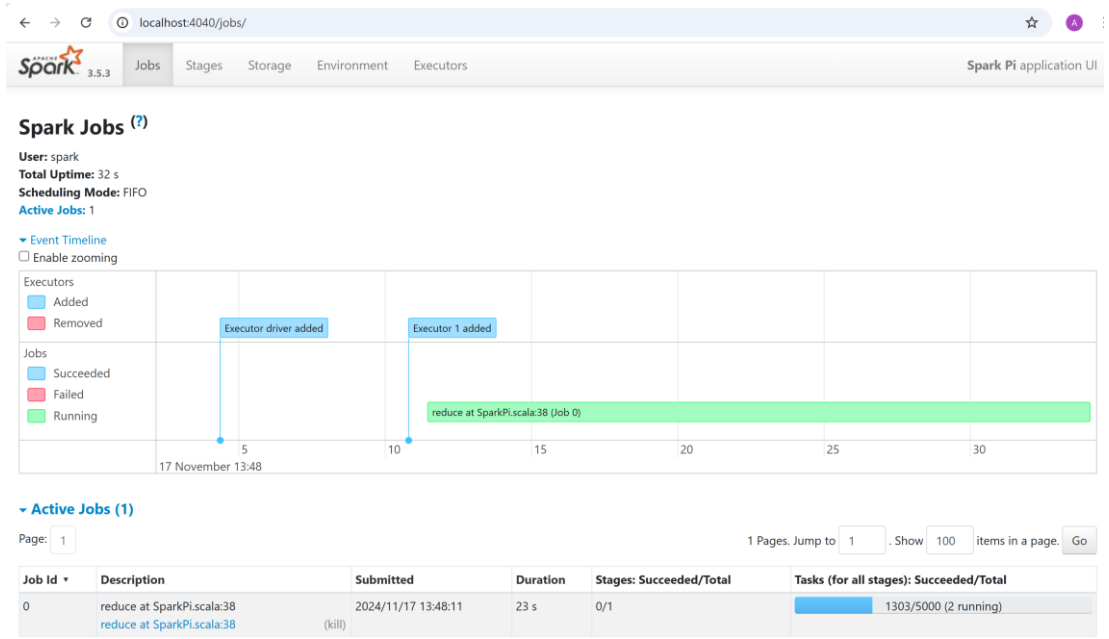


do "port forward" 4040 (in driver pod) -> 4040 on localhost

kubectl port-forward spark-pi-driver 4040:4040

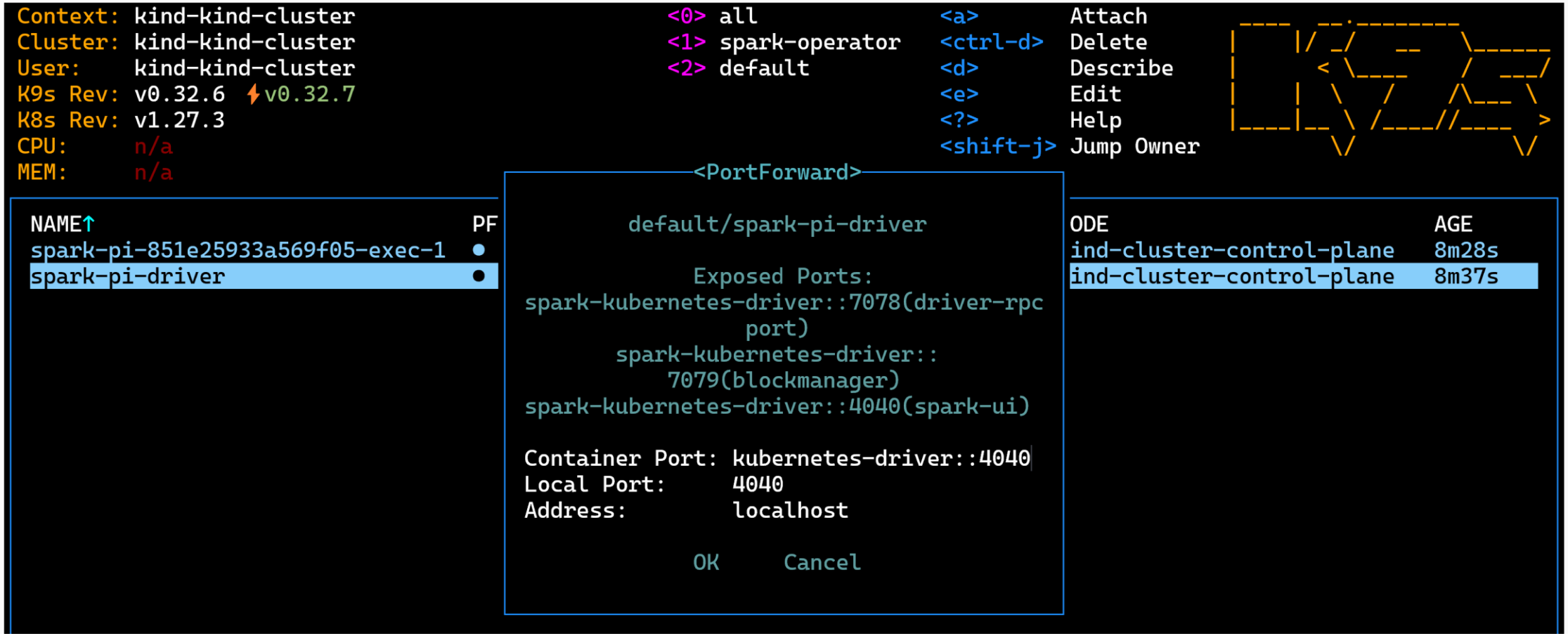
```
C:\apps\spark\spark-on-k8s>kubectl port-forward spark-pi-driver 4040:4040
Forwarding from 127.0.0.1:4040 -> 4040
Forwarding from [::1]:4040 -> 4040
```

Then open your browser on **htttp://localhost:4040**



Alternative: do port-forward using K9s

select pod "spark-pi-driver", type "<CONTROL> + F", edit port number "4040", then "OK"



It just works



Next Steps - Personnal Study

- try to relaunch several times, how to do?
- explore the internals of spark-operator + k8s pods
- change parameter to use several executors, and different cores number
- provide your own jar and className to execute
- execute a "shell" inside the spark-executor/driver to monitor unix process
- add environment variable to the driver/pod
- choose env variable _JAVA_OPTIONS to enable remote JVM debugging
  then attach debugger as "remote java process"
- use ConfigMap from k8s to be visible as mounted volume files
- use "emptyDir" mount volume for shuffle files
- etc etc

Dive Deeper,
Understand the Magie
& TroubleShoot Errors

running your first "Docker" spark image

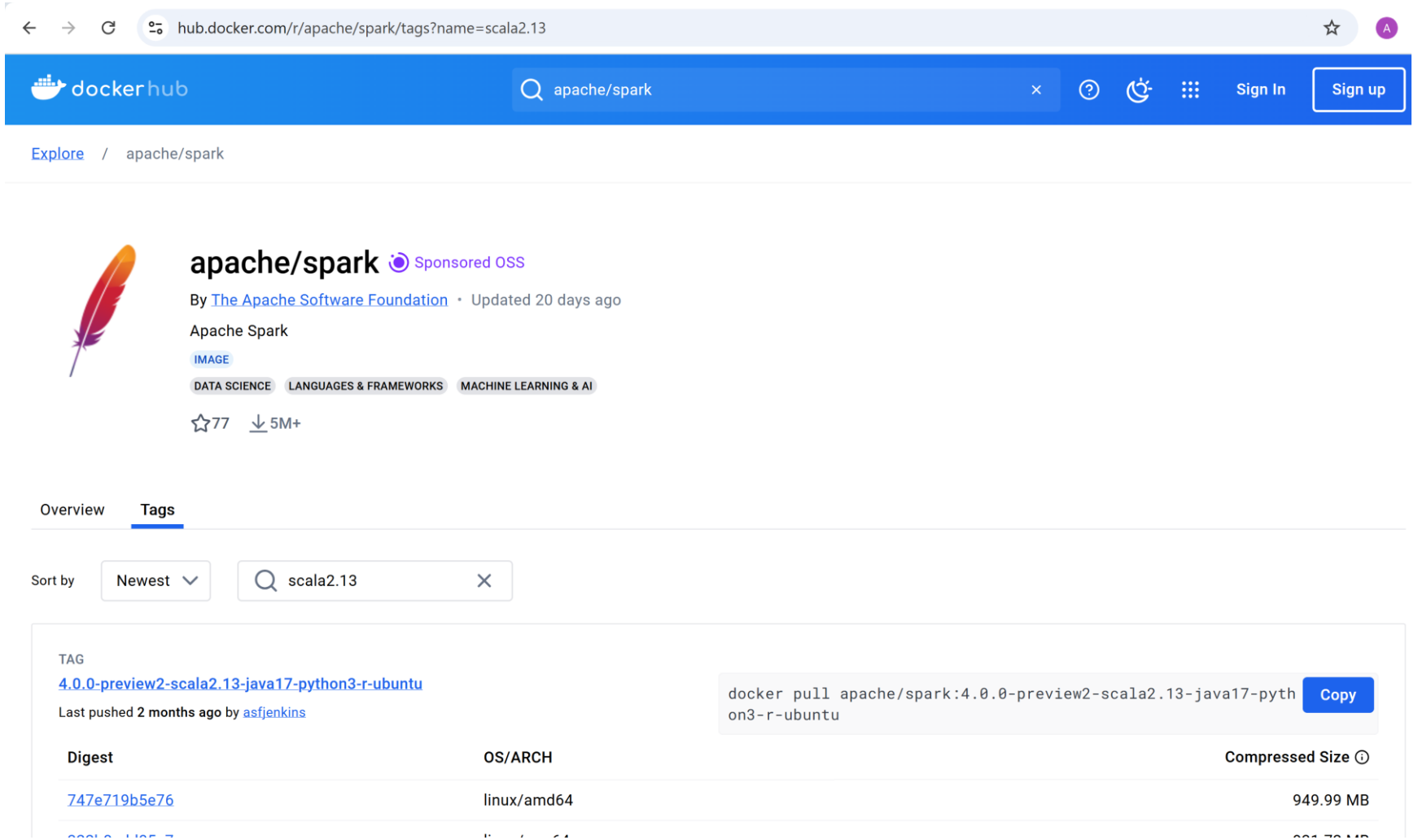
running spark image in Docker "local" mode

docker run -it apache/spark /opt/spark/bin/spark-shell

OR

podman run -it apache/spark /opt/spark/bin/spark-shell

Docker HUB spark "latest" image ? example: finding with scala2.13 (!= 2.12)



spark-shell --master local[*] in Docker

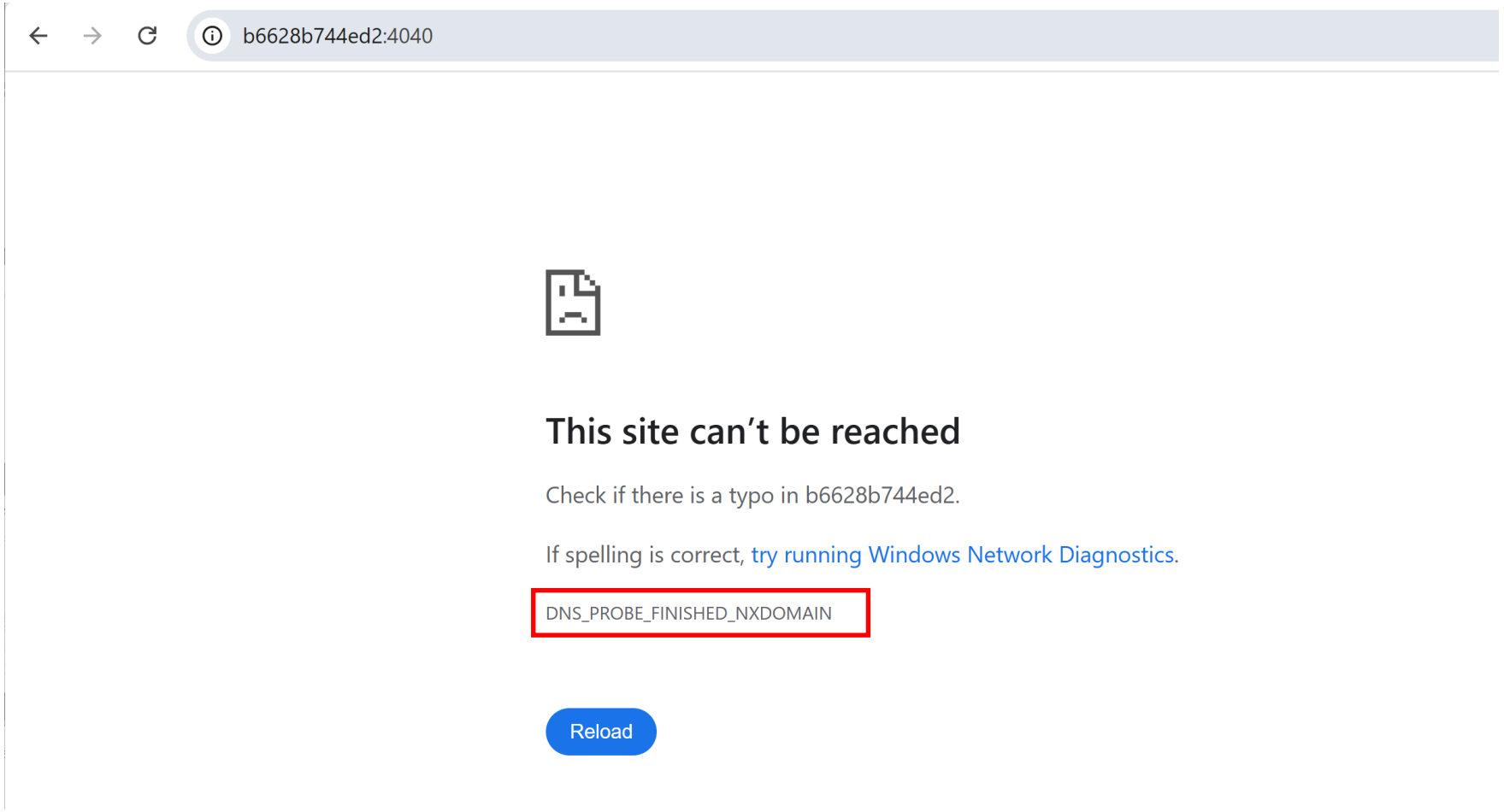
podman run -it apache/spark /opt/spark/bin/spark-shell

```
C:\Users\arnaud>podman run -it apache/spark /opt/spark/bin/spark-shell
Resolving "apache/spark" using unqualified-search registries (/etc/containers/registries.conf.d/999-podman-machine.conf)
Trying to pull docker.io/apache/spark:latest...
Getting image source signatures
Copying blob sha256:b3ef1f9e9d220aa4c081924e26f514bbafee5fecb9cb7f8e3b5b19762ce947fa
Copying blob sha256:65c79716ce2573e0cd3464a688211336d1ee703c3eef803ae4cf3438e0c64272
Copying blob sha256:560c024910bebac6b404791af28ebd48a8289303b8377d17b67ffdfe52754f2a
Copying blob sha256:e46bdbd88e5c97485003310b142bf5c3c58bb9677087aed67152e8d6e4d5f47a
Copying blob sha256:a3fac0216367b2e208f4a4bfc080dff215045e5e498a969d03cb155bed81049d
Copying blob sha256:9cffdbb20fcf65eb28ccf57ec92b97ed19233b4e2b7bd740b0fb41b6b8de2514
Copying blob sha256:507dcc9ae5b6139e01b1192a579d474e7e307add6c6bd0a77bd8e8db09ce0687
Copying blob sha256:0c93c619e2221a853dd59e90e774c4a36a6ff11a8de381ce12627ed474bb44ef
Copying blob sha256:de58e49cfa67a5774e2ffd22ddb044f4160620fa92f6b6364f39fe53a7209c68
Copying blob sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1
Copying blob sha256:2cadc3f91afb615794f14e6146477d532a01c4499a351fe7e8a97adde84704df
Copying config sha256:d3ea4aeb842bc149cfcb143f7692273c9077b84bef2942a43aeb36ac0b8169dc
Writing manifest to image destination
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
24/11/16 14:07:39 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Spark context Web UI available at http://b6628b744ed2:4040
Spark context available as 'sc' (master = local[*], app id = local-1731766061291).
Spark session available as 'spark'.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.5.3
      /_/

Using Scala version 2.12.18 (OpenJDK 64-Bit Server VM, Java 11.0.24)
Type in expressions to have them evaluated.
Type :help for more information.

scala>
```

open SparkUI ? localhost ? temporary "vm name" not found



Need port-forward

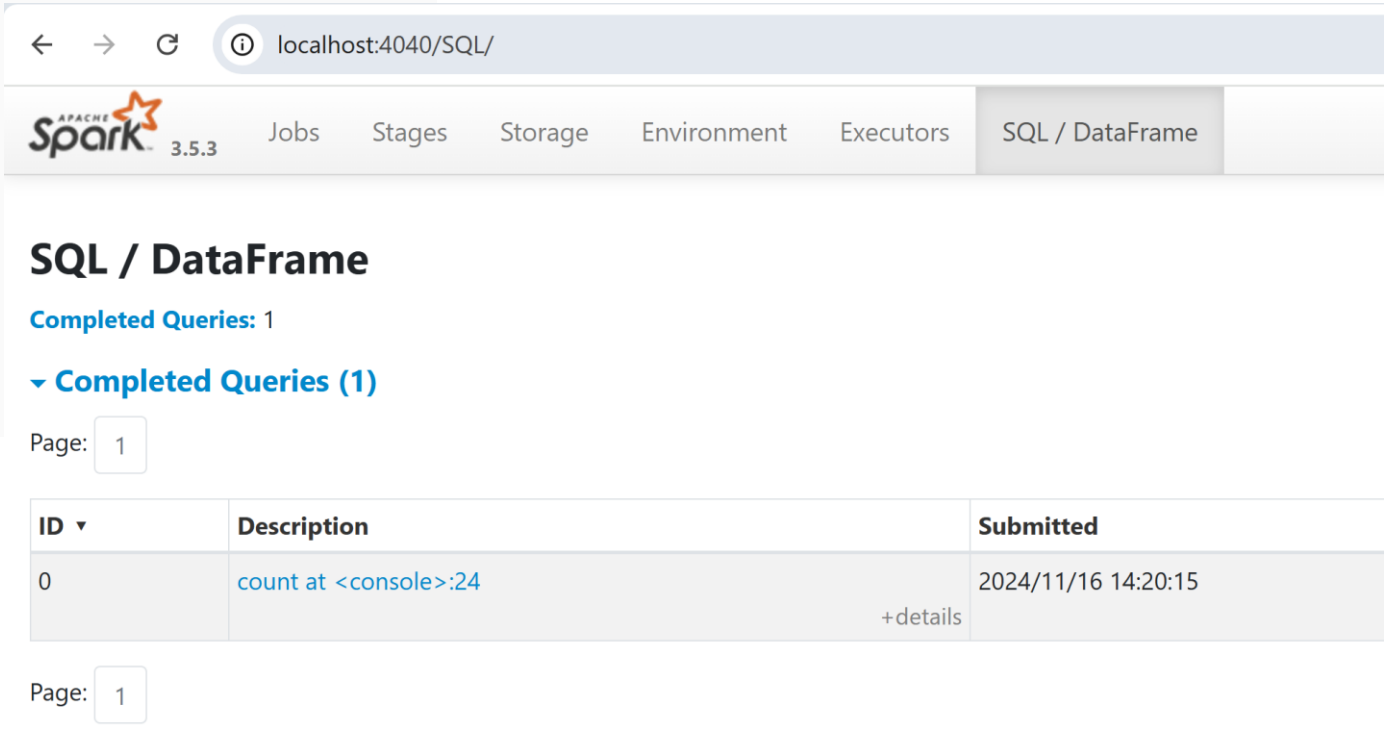
podman run -it -p 4040:4040 apache/spark /opt/spark/bin/spark-shell

```
C:\Users\arnaud>podman run -it -p 4040:4040 apache/spark /opt/spark/bin/spark-shell
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
24/11/16 14:18:43 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform
Spark context Web UI available at http://f9dfeec3f1a0:4040
Spark context available as 'sc' (master = local[*], app id = local-1731766724994).
Spark session available as 'spark'.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.5.3
      /_/

Using Scala version 2.12.18 (OpenJDK 64-Bit Server VM, Java 11.0.24)
Type in expressions to have them evaluated.
Type :help for more information.

scala> val ds=spark.createDataset(Seq(1))
ds: org.apache.spark.sql.Dataset[Int] = [value: int]

scala> ds.count()
res0: Long = 1
```



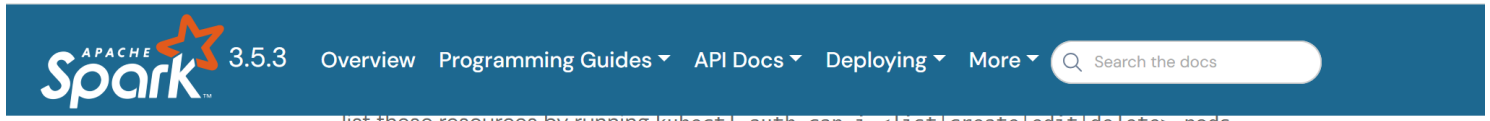
ok... running "docker java" instead of "java"

but not yet using "--master k8s://k8s-api-server"

still using "--master local[*]"

built-in in Spark (java/scala) code : --master k8s://

https://spark.apache.org/docs/3.5.3/running-on-kubernetes.html

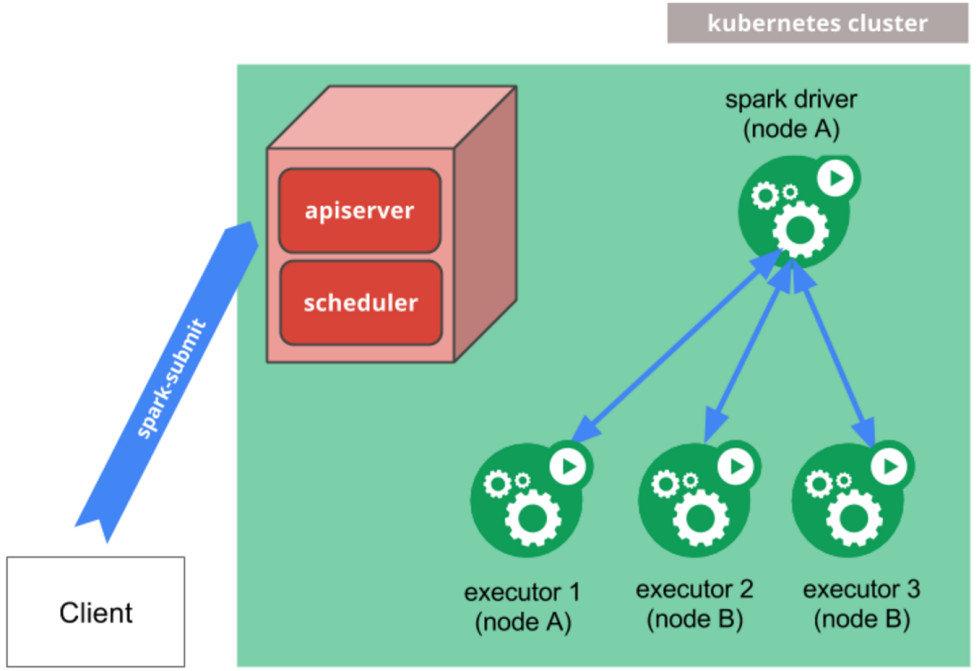


list these resources by running `kubectl auth can-i <list|create|edit|delete> pods`.

- The service account credentials used by the driver pods must be allowed to create pods, services and configmaps.

- You must have Kubernetes DNS configured in your cluster.

How it works



kubectl cluster-info
=> local k8s Api server : port

```
$ kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:53779
CoreDNS is running at https://127.0.0.1:53779/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
```

doc example
--master k8s://localhost:port

```
spark-submit \
    --master k8s://https://localhost:port \
    --deploy-mode cluster \
    --name spark-pi \
    --class org.apache.spark.examples.SparkPi \
    --conf spark.kubernetes.container.image=apache/spark:latest \
    local:///path/to/examples.jar
```

Command ... for local spark

spark-submit --master k8s://localhost:53779 --deploy-mode cluster --name spark-pi --class org.apache.spark.examples.SparkPi --conf spark.kubernetes.container.image=apache/spark:4.0.0-preview2-scala2.13-java17-python3-r-ubuntu local://examples/jars/spark-examples_2.13-3.5.0.jar

INVALID ... on purpose (can skip question, or investigate)

```
C:\apps\spark\spark-3.5.0>dir examples\jars
 Le volume dans le lecteur C s'appelle Windows-SSD
 Le numéro de série du volume est 3EFF-0F82

 Répertoire de C:\apps\spark\spark-3.5.0\examples\jars

09/09/2023  02:40    <DIR>          .
09/09/2023  02:40    <DIR>          ..
09/09/2023  02:40            80 424 scopt_2.13-3.7.1.jar
09/09/2023  02:40         1 588 886 spark-examples_2.13-3.5.0.jar
               2 fichier(s)        1 669 310 octets
               2 Rép(s)  176 976 617 472 octets libres

C:\apps\spark\spark-3.5.0>spark-submit --master k8s://localhost:53779 --deploy-mode cluster --name spark-pi --class org.apache.spark.examples.SparkPi --conf spark
.kubernetes.container.image=apache/spark:4.0.0-preview2-scala2.13-java17-python3-r-ubuntu local://examples/jars/spark-examples_2.13-3.5.0.jar
```

... PENDING ???

Pending few minutes (intermediate Warning "MountVolume failed" will be auto resolved with wait+internl retry)

```
Context: kind-kind-cluster                    <c>       Copy
Cluster: kind-kind-cluster                    <e>       Edit
User:    kind-kind-cluster                    <n>       Next Match
K9s Rev: v0.32.6                              <shift-n> Prev Match
K8s Rev: v1.27.3                              <r>       Toggle Auto-Refresh
CPU:     n/a                                  <f>       Toggle FullScreen
MEM:     n/a
                                                                  ____  __.________
                                                                 |    |/ _/   __   \_____
                                                                 |      < \____    /  ___/
                                                                 |    |  \    /    /\___ \
                                                                 |____|__ \ /____//____  >
                                                                         \/           \/
                    Describe(default/spark-pi-2b5a8b933579ffda-driver)
    Type:                       Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:     3607
    ConfigMapName:              kube-root-ca.crt
    ConfigMapOptional:          <nil>
    DownwardAPI:                true
QoS Class:                      Burstable
Node-Selectors:                 <none>
Tolerations:                    node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                                node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:
  Type     Reason       Age    From               Message
  ----     ------       ----   ----               -------
  Normal   Scheduled    2m24s  default-scheduler  Successfully assigned default/spark-pi-2b5a8b933579ffda-driver to
kind-cluster-control-plane
  Warning  FailedMount  2m25s  kubelet            MountVolume.SetUp failed for volume "spark-conf-volume-driver" : c
onfigmap "spark-drv-17209a93357a04a6-conf-map" not found
  Normal   Pulling      2m24s  kubelet            Pulling image "apache/spark:4.0.0-preview2-scala2.13-java17-python
3-r-ubuntu"


<pod>  <describe>
```

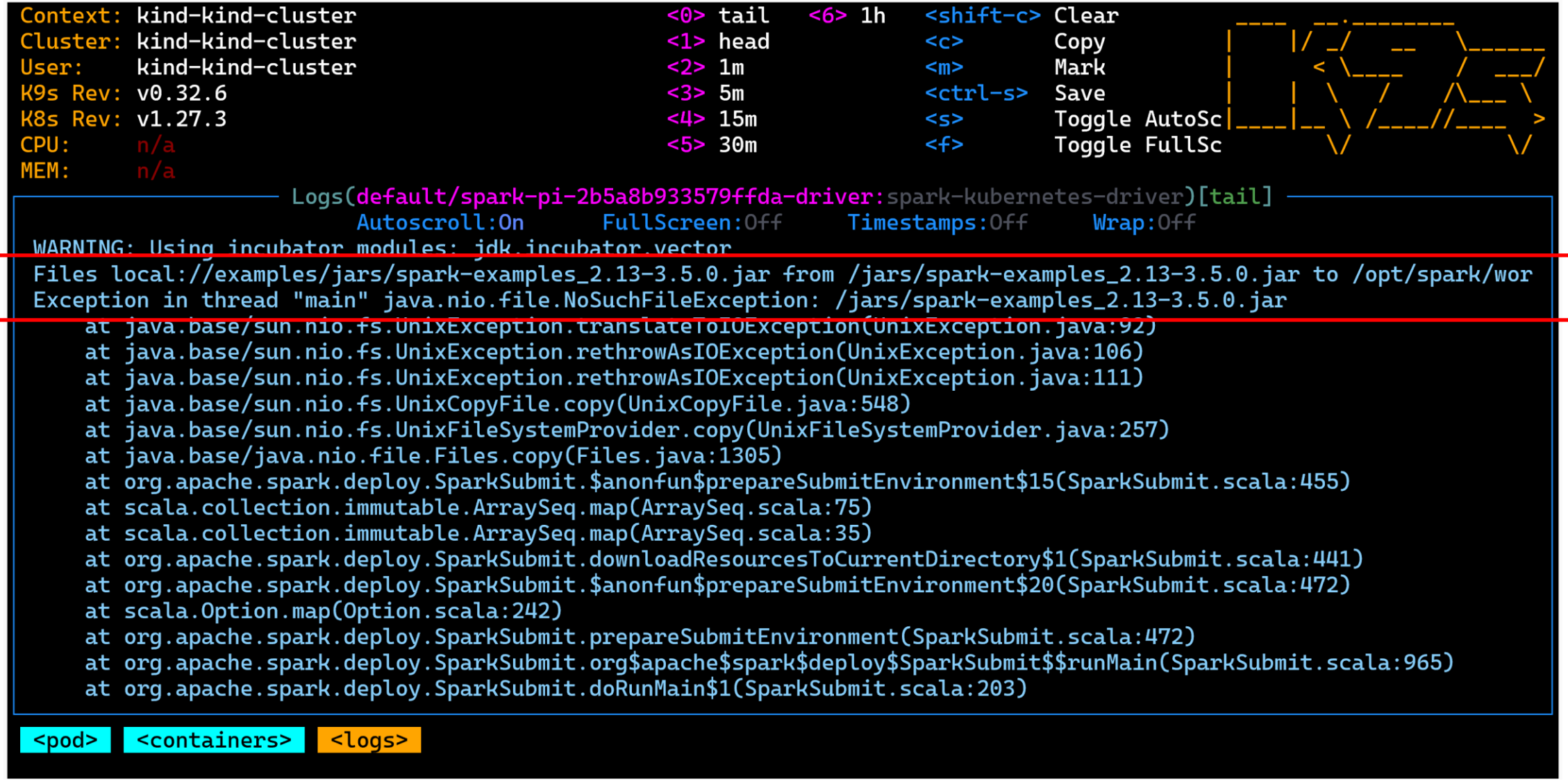
Diagnosing Error : K9s - pod

```
Context: kind-kind-cluster                <0> all              <a>       Attach
Cluster: kind-kind-cluster                <1> spark-operator   <ctrl-d>  Delete
User:    kind-kind-cluster                <2> default          <d>       Describe
K9s Rev: v0.32.6                                              <e>       Edit
K8s Rev: v1.27.3                                              <?>       Help
CPU:     n/a                                                  <shift-j> Jump Owner
MEM:     n/a
                                  Pods(default)[1]
NAME↑                                PF  READY  STATUS  RESTARTS IP           NODE                          AGE
spark-pi-2b5a8b933579ffda-driver     ●   0/1    Error          0 10.244.0.17   kind-cluster-control-plane    7m21s
```

K9s pod "driver" -> container "spark-kubernetes-driver"

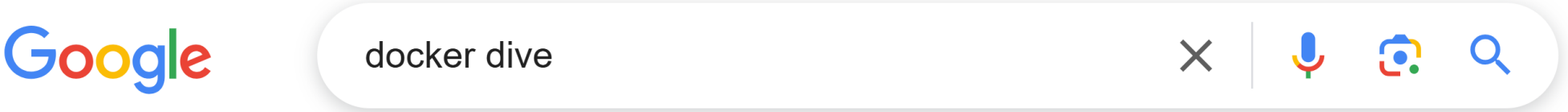
```
Context: kind-kind-cluster                <a>       Attach          <f> Show PortFor…
Cluster: kind-kind-cluster                <?>       Help
User:    kind-kind-cluster                <l>       Logs
K9s Rev: v0.32.6                          <p>       Logs Previous
K8s Rev: v1.27.3                          <shift-f> PortForward
CPU:     n/a                              <s>       Shell
MEM:     n/a
                        Containers(default/spark-pi-2b5a8b933579ffda-driver)[1]
IDX↑  NAME                    PF IMAGE                                                READY  STATE  TARTS
M1    spark-kubernetes-driver   ● apache/spark:4.0.0-preview2-scala2.13-java17-python3-r-ubuntu  false  Error      0
```

k9s pod - container - log
… jar file not found



"local://" means a file already existing "inside" the image !

how to explore existing Docker Image content ? "dive"



All Images Videos News Web Books Finance Tools

GitHub
https://github.com › wagoodman › dive

wagoodman/dive: A tool for exploring each layer in ...

A tool for exploring a **docker** image, layer contents, and discovering ways to shrink the size of your **Docker**/OCI image.

Dockerfile · README.md · Issues 169 · Pull requests 37

docker dive <<image>>

github.com/wagoodman/dive

README MIT license



To analyze a Docker image simply run dive with an image tag/id/digest:

```
dive <your-image-tag>
```

or you can dive with docker command directly

```
alias dive="docker run -ti --rm  -v /var/run/docker.sock:/var/run/docker.sock wagoodman/dive"
dive <your-image-tag>
```

```
# for example
dive nginx:latest
```

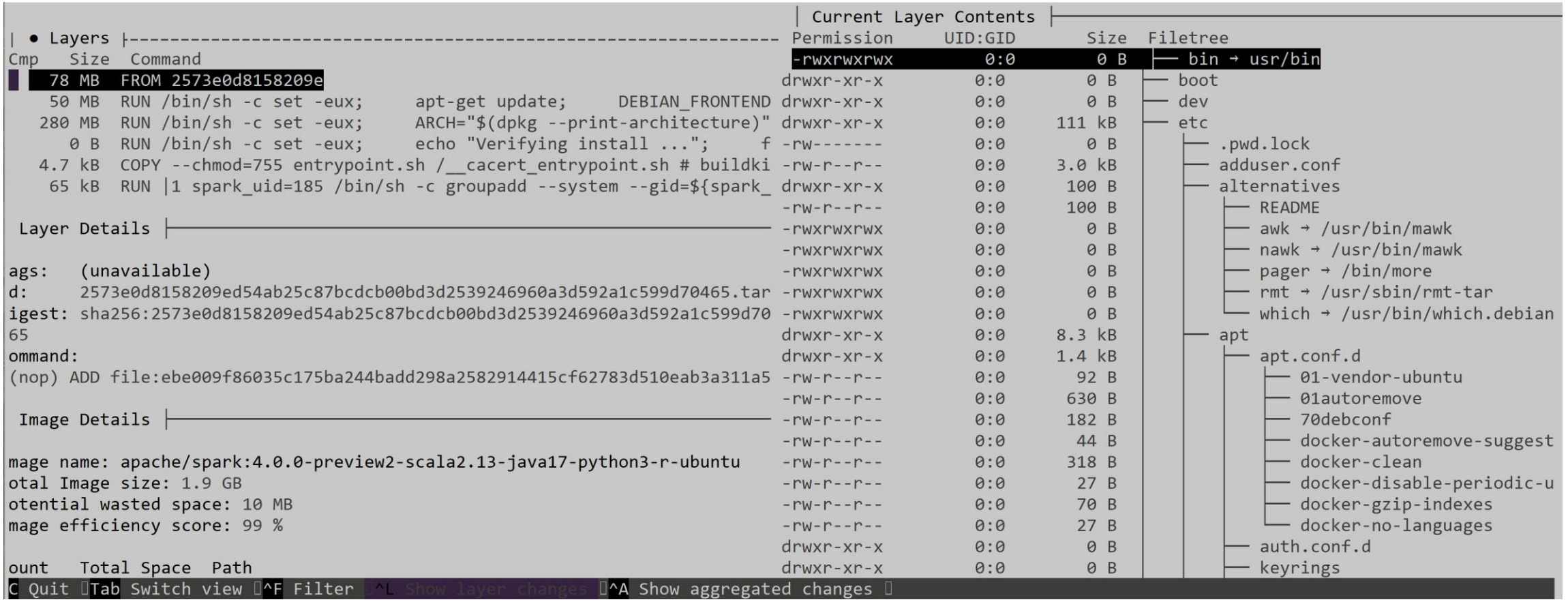
podman dive run -it wagoodman/dive <<img>>

podman run -ti --rm -v /var/run/docker.sock:/var/run/docker.sock wagoodman/dive apache/spark:4.0.0-preview2-scala2.13-java17-python3-r-ubuntu

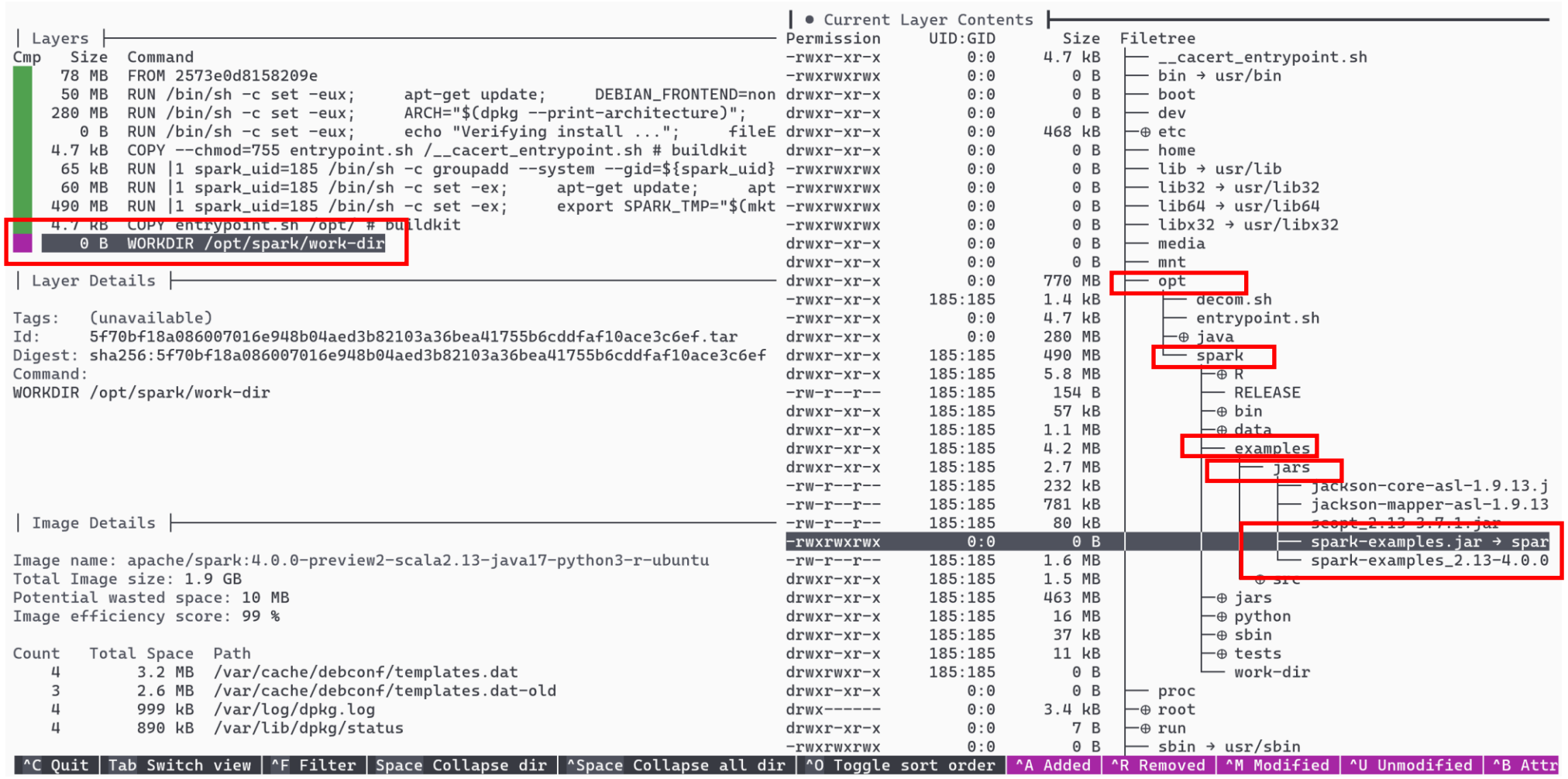
```
$ alias dive="podman run -ti --rm  -v /var/run/docker.sock:/var/run/docker.sock wagoodman/dive"

arnaud@DesktopArnaud /cygdrive/c/apps/hadoop
$ dive apache/spark:4.0.0-preview2-scala2.13-java17-python3-r-ubuntu
Resolving "wagoodman/dive" using unqualified-search registries (/etc/containers/registries.conf.d/999-podman-machine.conf)
Trying to pull docker.io/wagoodman/dive:latest...
Getting image source signatures
Copying blob sha256:a7dfad62b2dab9e4aae65fd8fc6682e8841a6876a31a3675495f7554e93b0f46
Copying blob sha256:cf6c0733f07b22c9a8c341023181bdb5f273c050d416bd4452bfd96589c4a5d0
Copying blob sha256:619be1103602d98e1963557998c954c892b3872986c27365e9f651f5bc27cab8
Copying config sha256:cf2f0c72f08544080eaf06781133b5943bace97a55d8f8fb61b32e6051712245
Writing manifest to image destination
Image Source: docker://apache/spark:4.0.0-preview2-scala2.13-java17-python3-r-ubuntu
Fetching image... (this can take a while for large images)
```

dive - browse layers



Navigate: on Layer step, then "->" for right pane then "<SPACE>" for collapse dir



alternative:

docker run -it <<image>> "/bin/bash"

(if image contains a shell)

then # cd .. ; ls ; find

docker run -it

```
C:\apps\spark\spark-3.5.0>podman run -it apache/spark:4.0.0-preview2-scala2.13-java17-python3-r-ubuntu /bin/bash
Resolving "apache/spark" using unqualified-search registries (/etc/containers/registries.conf.d/999-podman-machine.conf)
Trying to pull docker.io/apache/spark:4.0.0-preview2-scala2.13-java17-python3-r-ubuntu...
Getting image source signatures
Copying blob sha256:7ad3b5e04f2c57070106a37d5d44429d29123ae69ff54b93f63c101129c298d9
Copying blob sha256:7478e0ac0f23f94b2f27848fbcdf804a670fbf8d4bab26df842d40a10cd33059
Copying blob sha256:80584b9b4ef704ac9fe073b939107337aa86c0c2b9d7742d863bdb8f219a9463
Copying blob sha256:589e94d8e4bef4afb4cff24dcb435416fa3f42401c7ab9b0401e913516968373
Copying blob sha256:21beeebd32c0abccb0b45e99a7fd283212bb21ac4fad10c36216455082df95d0
Copying blob sha256:fcbf49cabfbfcba18d6177c1ac6c7e9330c15d3d20cbc4f88c58bd335ee450a2
Copying blob sha256:b0775d58952aef063d10d90c24833664221c93bbb444d3543830357deb36d14b
Copying blob sha256:519ab62359f396e5daecd01786e2a1ee9c3009be34b87fc5bf66328989738983
Copying blob sha256:7978e188866e12963b2cf829d0b1d08409003abbe7f0ea466622e34e53d34130
Copying blob sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1
Copying blob sha256:b0aad1ccdf6835e7740225bcb7acd0547516678a5648cabff50ed5656d8b12cd
Copying config sha256:ea79fc05d218a268153c4cd35fa3d306965c23c59232759d25aeb4862accbca7
Writing manifest to image destination

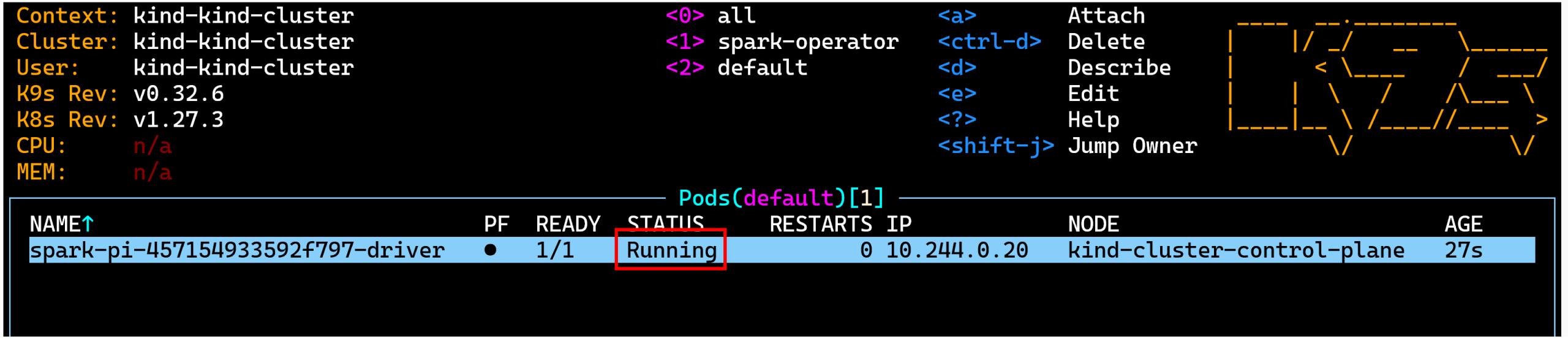
spark@4c64f3782002:/opt/spark/work-dir$
spark@4c64f3782002:/opt/spark/work-dir$ ls /opt/
decom.sh       entrypoint.sh  java/          spark/
spark@4c64f3782002:/opt/spark/work-dir$ ls /opt/spark/examples/
jars/ src/
spark@4c64f3782002:/opt/spark/work-dir$ ls /opt/spark/examples/jars/
jackson-core-asl-1.9.13.jar             scopt_2.13-3.7.1.jar                    spark-examples.jar
jackson-mapper-asl-1.9.13.jar           spark-examples_2.13-4.0.0-preview2.jar
spark@4c64f3782002:/opt/spark/work-dir$ ls /opt/spark/examples/jars/
jackson-core-asl-1.9.13.jar  jackson-mapper-asl-1.9.13.jar  scopt_2.13-3.7.1.jar  spark-examples_2.13-4.0.0-preview2.jar  spark-examples.jar
spark@4c64f3782002:/opt/spark/work-dir$
```



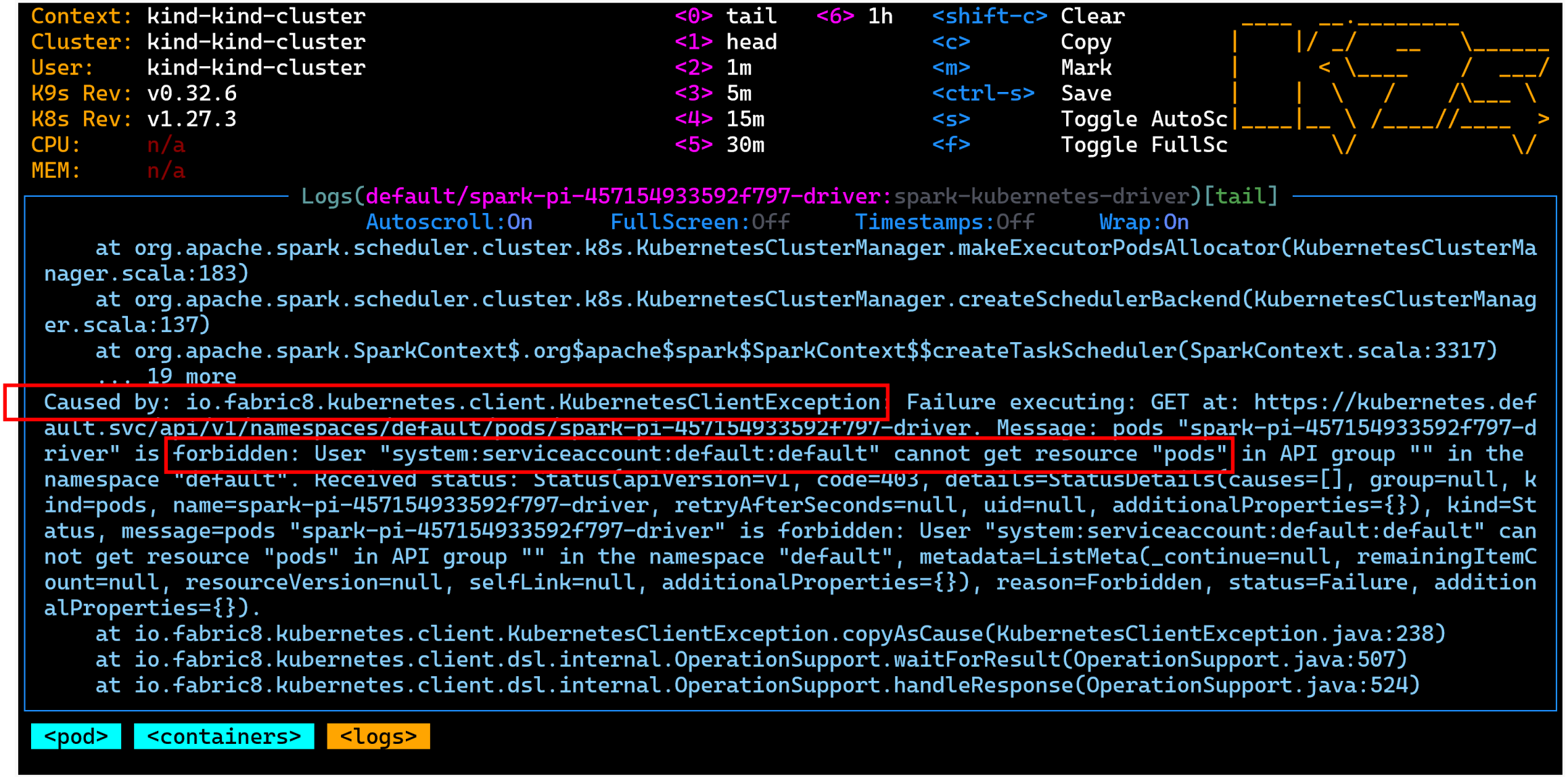
like "SSH" inside the pod container !

Relaunch => pod Running "spark-driver" but Failed to launch "spark-executor"

spark-submit --master k8s://localhost:53779 --deploy-mode cluster --driver-memory 1g --executor-memory 1g
--name spark-pi --class org.apache.spark.examples.SparkPi
--conf spark.kubernetes.container.image=apache/spark:4.0.0-preview2-scala2.13-java17-python3-r-ubuntu
local:///opt/spark/examples/jars/spark-examples.jar

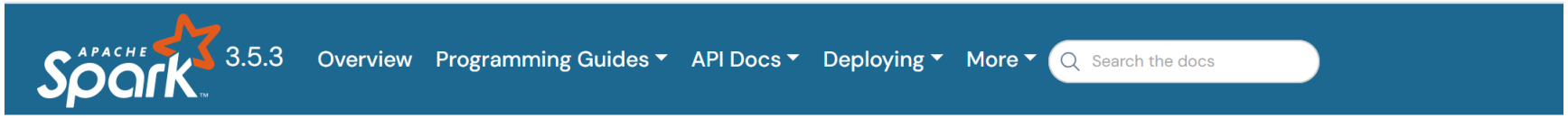


Failure : forbidden: User "...default" can not get resource "pods"



Doc serviceAccountName

https://spark.apache.org/docs/3.5.3/running-on-kubernetes.html



pods. The service account used by the driver pod must have the appropriate permission for the driver to be able to do its work. Specifically, at minimum, the service account must be granted a `Role` or `ClusterRole` that allows driver pods to create pods and services. By default, the driver pod is automatically assigned the `default` service account in the namespace specified by `spark.kubernetes.namespace`, if no service account is specified when the pod gets created.

Depending on the version and setup of Kubernetes deployed, this `default` service account may or may not have the role that allows driver pods to create pods and services under the default Kubernetes RBAC policies. Sometimes users may need to specify a custom service account that has the right role granted. Spark on Kubernetes supports specifying a custom service account to be used by the driver pod through the configuration property `spark.kubernetes.authenticate.driver.serviceAccountName=<service account name>`. For example, to make the driver pod use the `spark` service account, a user simply adds the following option to the `spark-submit` command:

```
--conf spark.kubernetes.authenticate.driver.serviceAccountName=spark
```

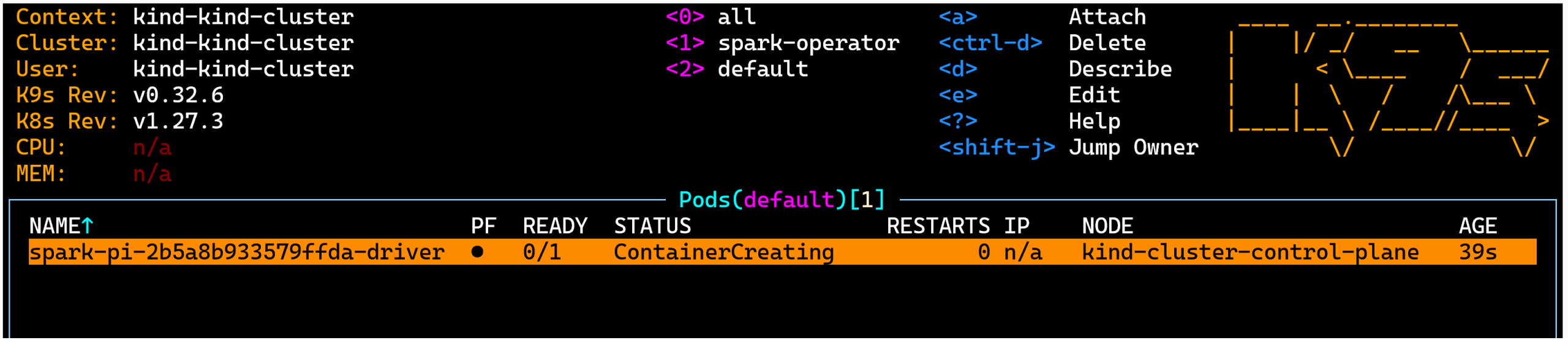
To create a custom service account, a user can use the `kubectl create serviceaccount` command. For example, the following command creates a service account named `spark`:

```
$ kubectl create serviceaccount spark
```

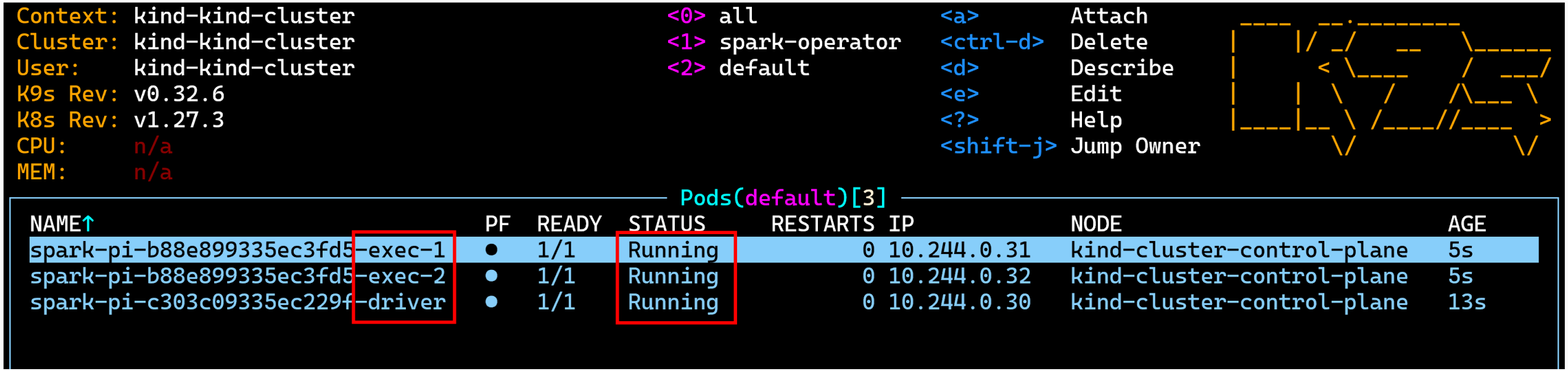
To grant a service account a `Role` or `ClusterRole`, a `RoleBinding` or `ClusterRoleBinding` is needed. To create a `RoleBinding` or `ClusterRoleBinding`, a user can use the `kubectl create rolebinding` (or `clusterrolebinding` for `ClusterRoleBinding`) command. For example, the following command creates an `edit` `ClusterRole` in the `default` namespace and grants it to the `spark` service account created above:

```
$ kubectl create clusterrolebinding spark-role --clusterrole=edit --serviceaccount=default:spark --
namespace=default
```

Relaunched: K9s pod - ContainerCreating

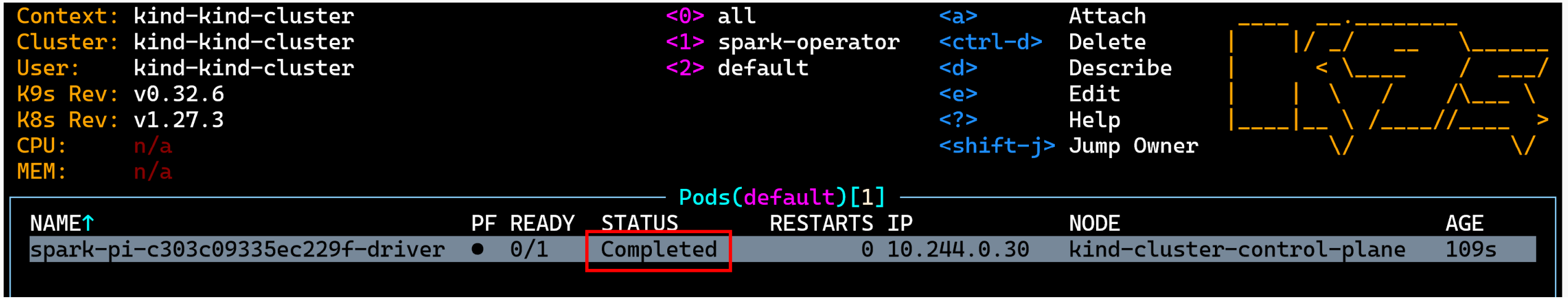


Relaunched: -> Running

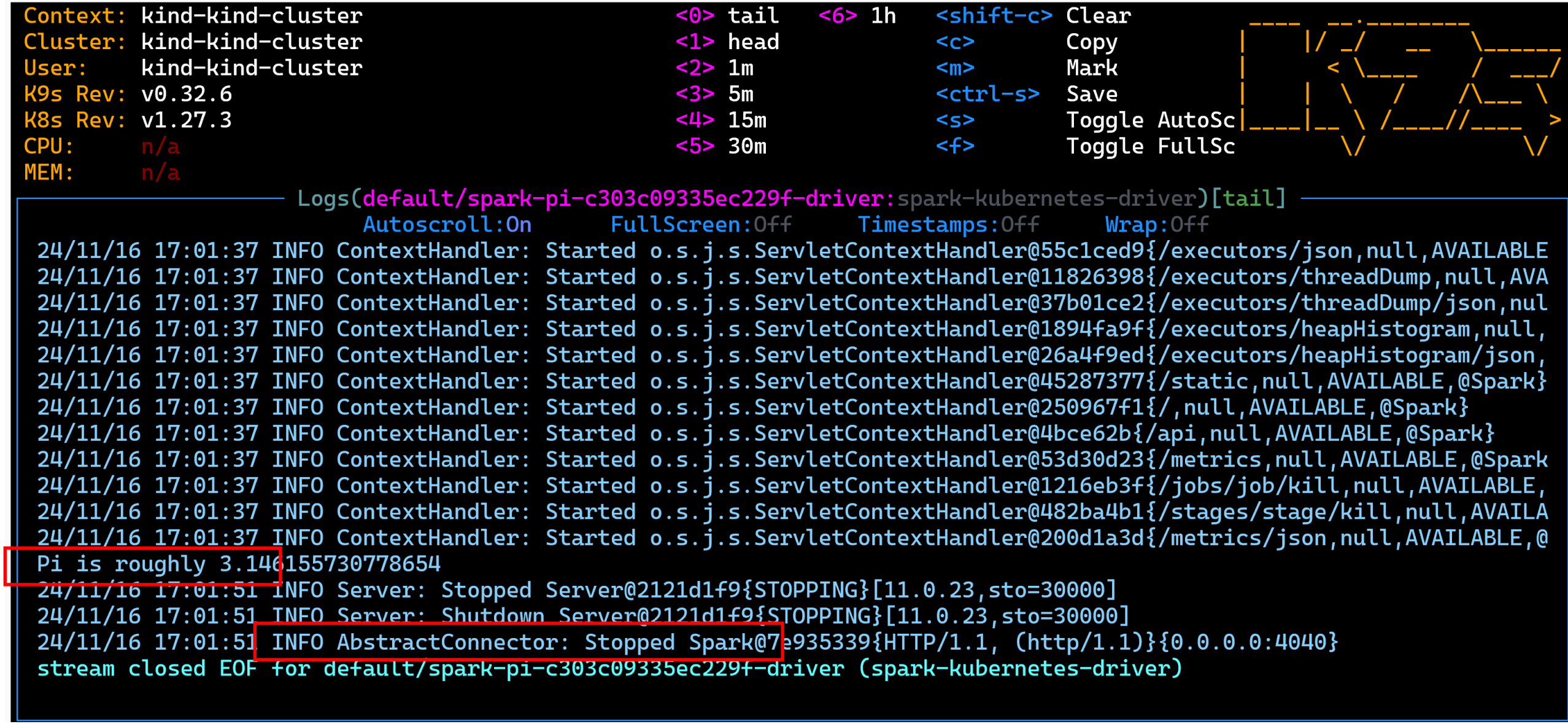


Notice : 3 pods = 1 driver + 2 * executors

Relaunched .. Completed



Completed - Logs



Scenario
if/when requesting too much memory

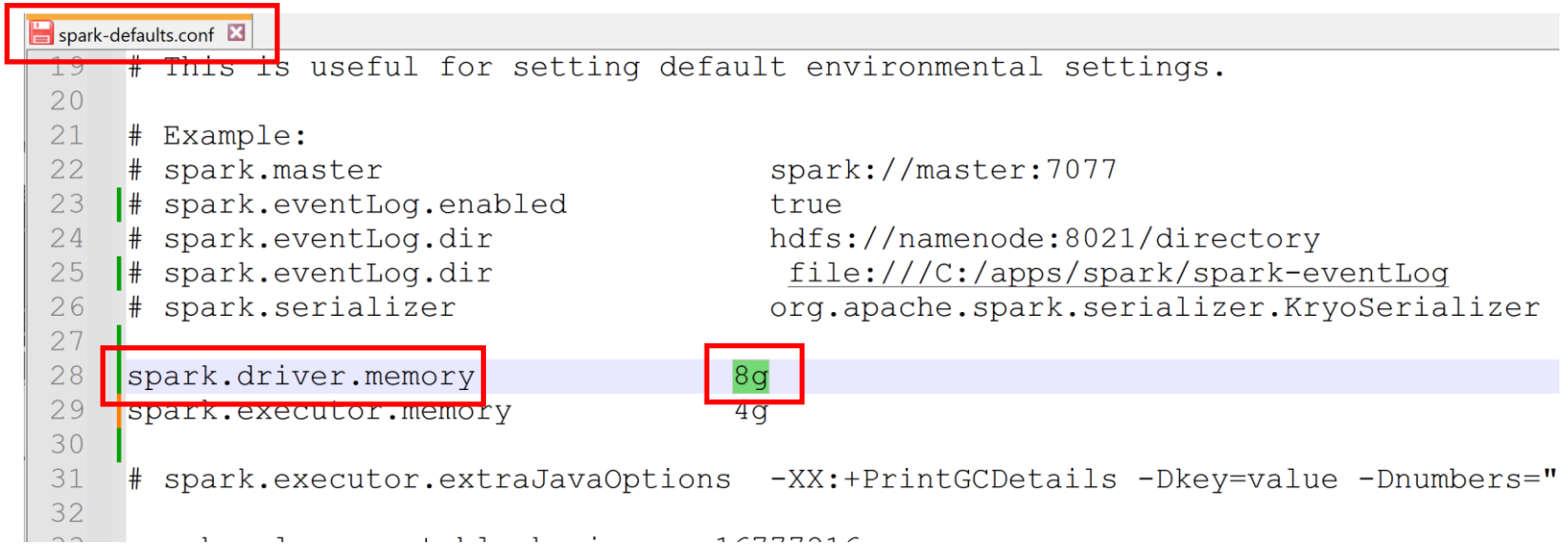
spark-submit

--driver-memory 8g

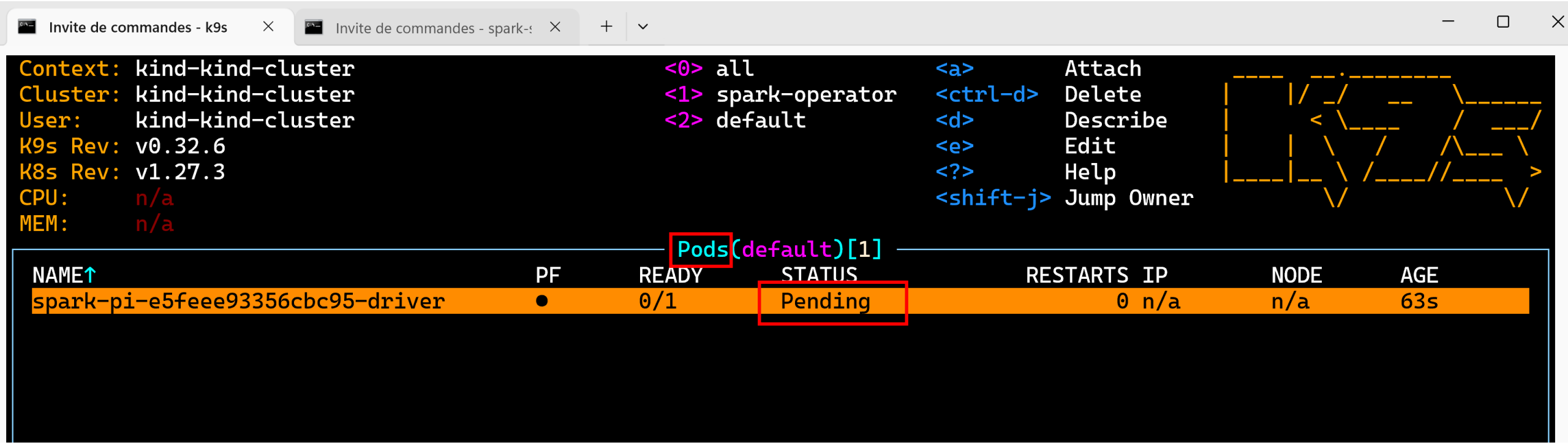
...

OR

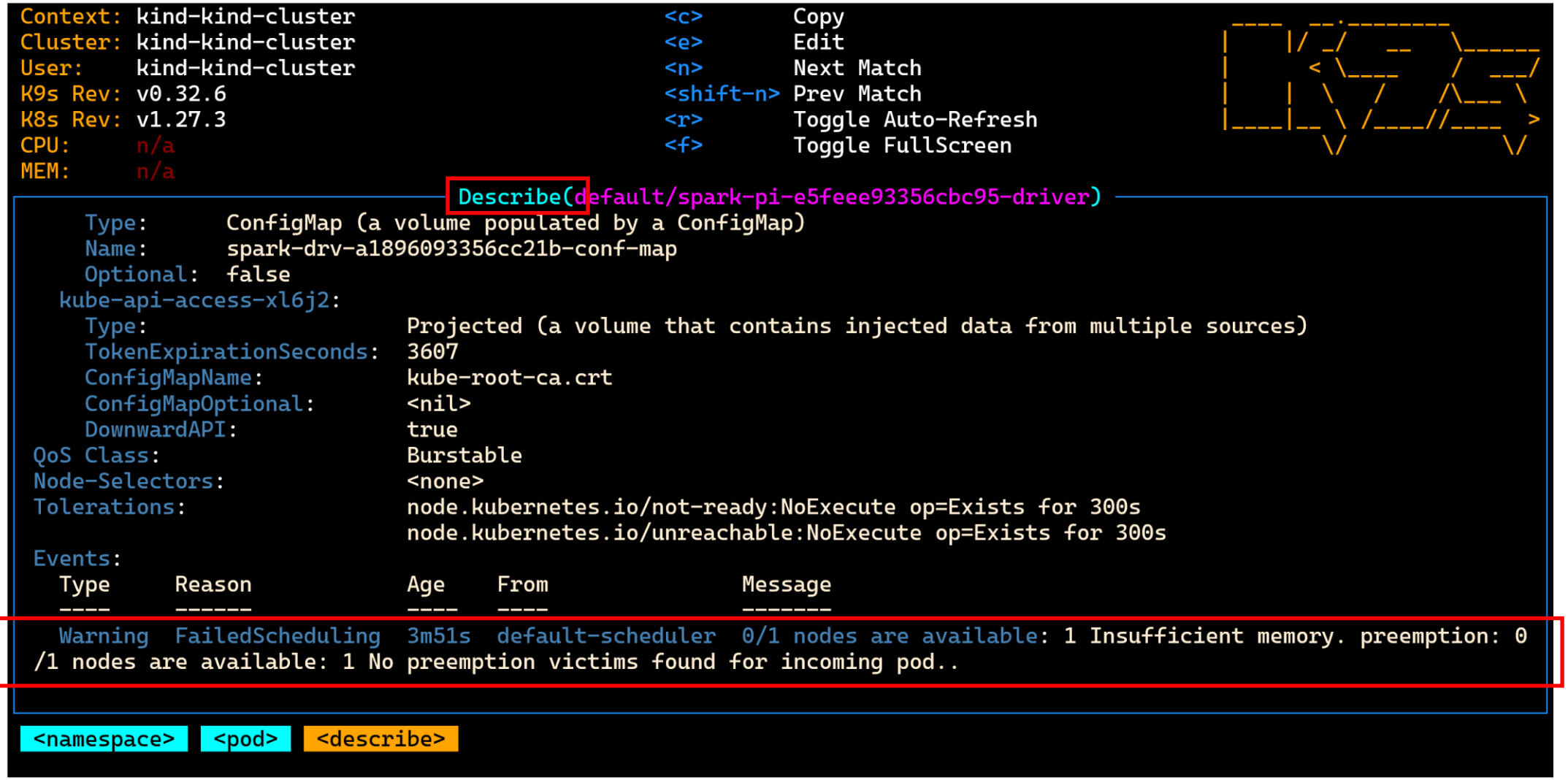
config file ${SPARK_HOME}/conf/spark-default.conf



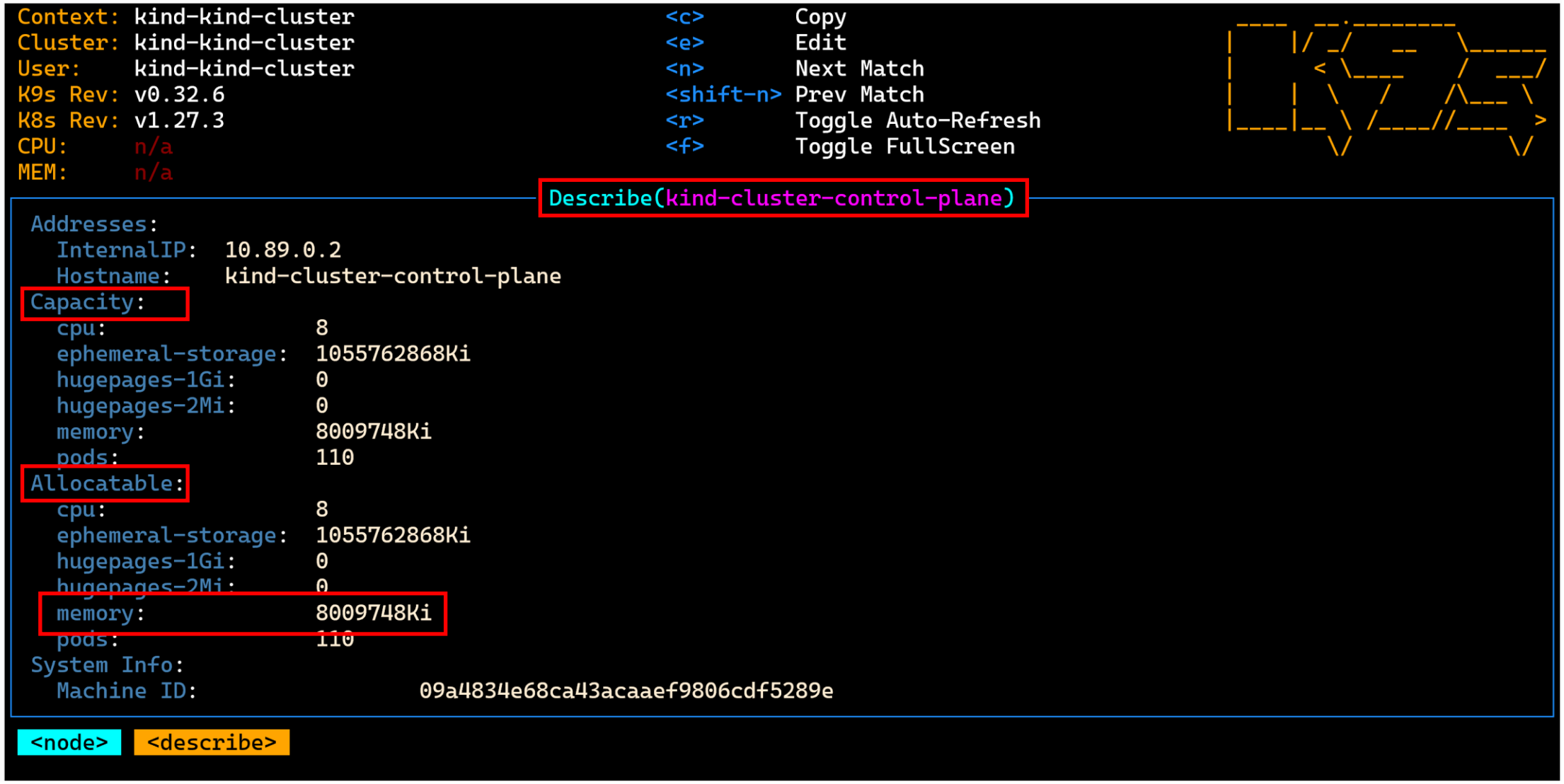
check in k9s (Pending Forever)



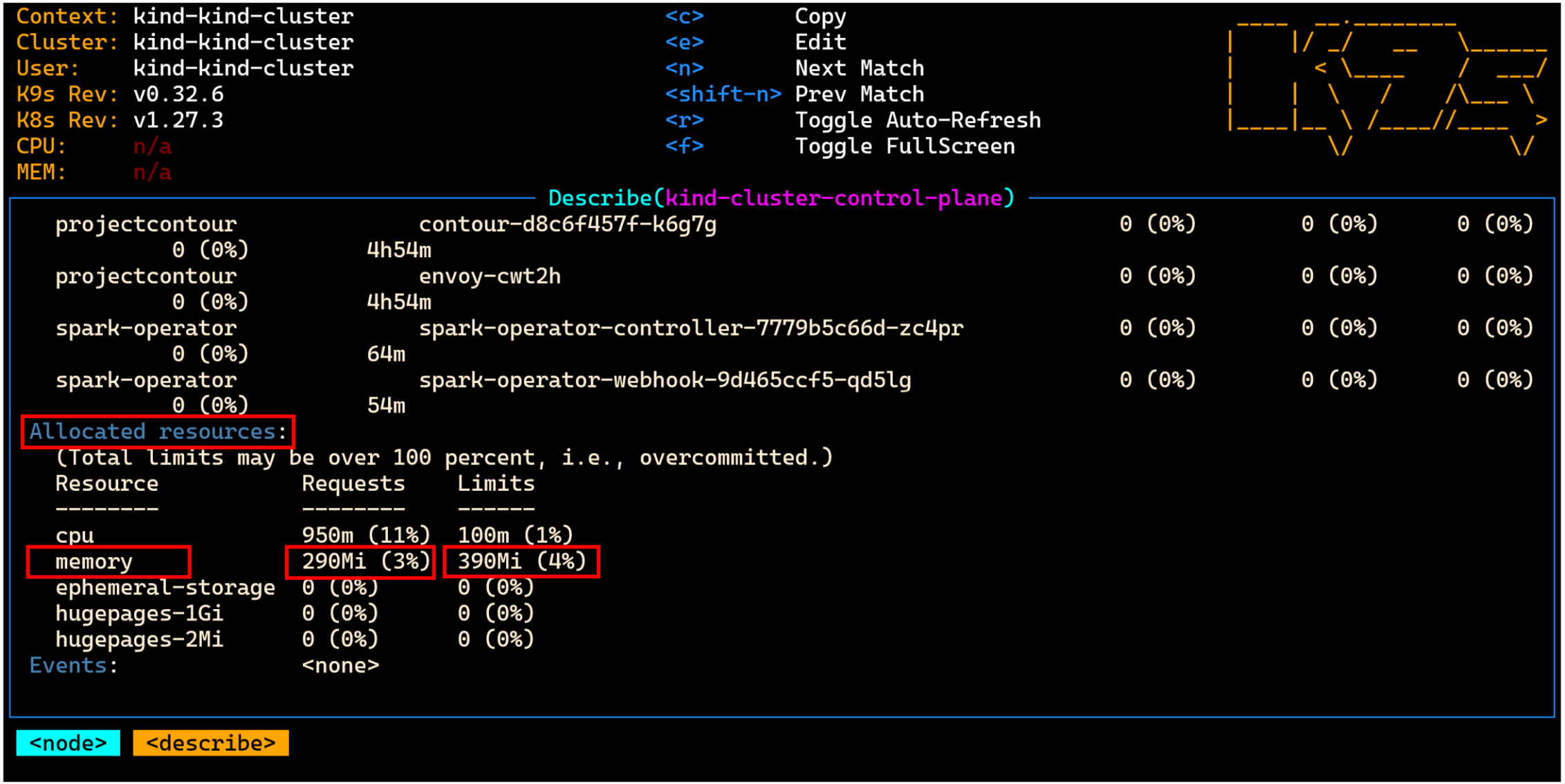
Debug "Pending" reason in K9s pod : "Insufficient memory"



K9s node : Capacity -> Allocatable -> Free Memory?



k9s node - Allocated / Free memory



Looks "acceptable" to run "--master k8s://"

BUT it is NOT practical:
no configMap created, no PVC, no Ingress, no Service..

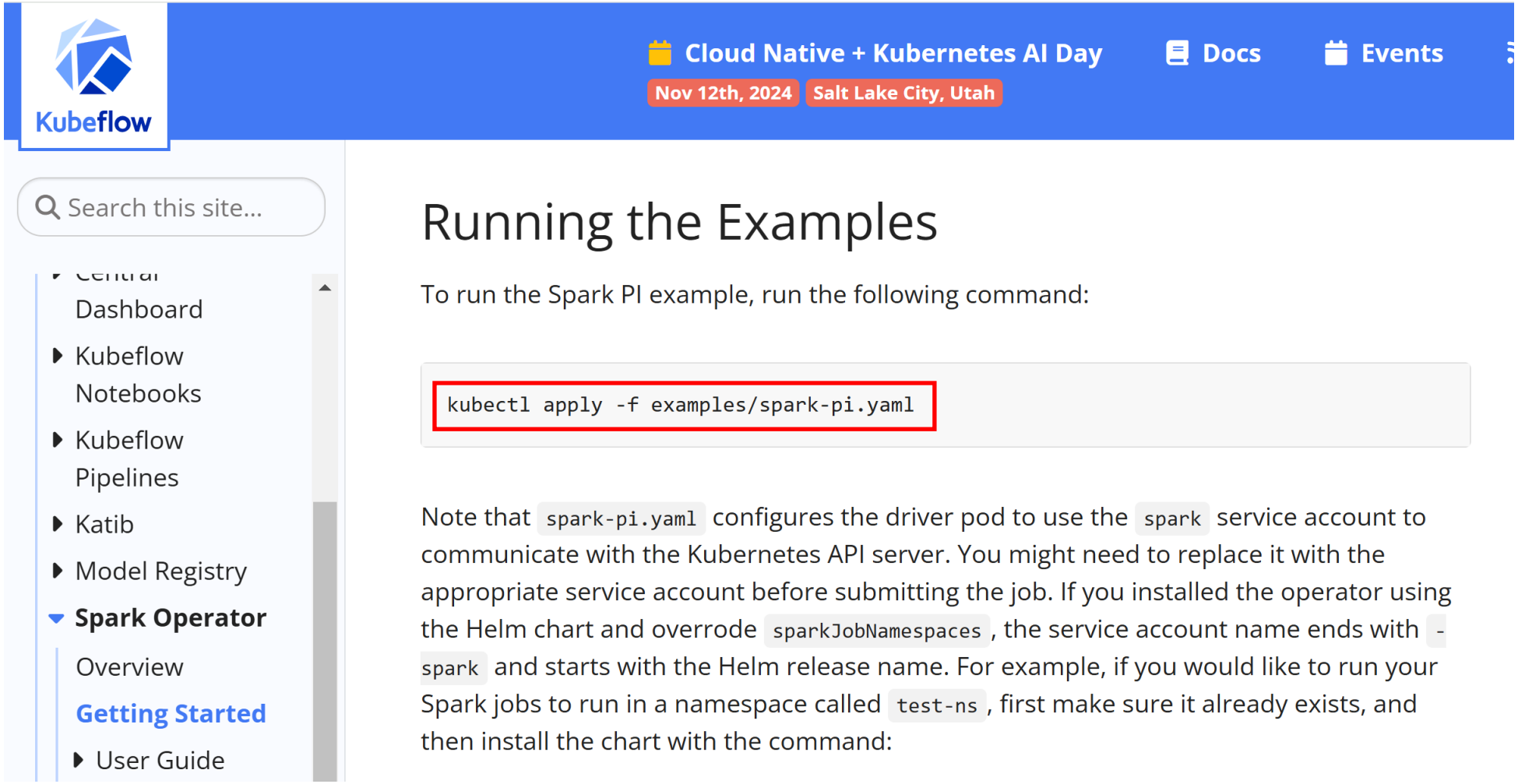
=> not sufficiently exposed/integrated in k8s

Time to launch for real

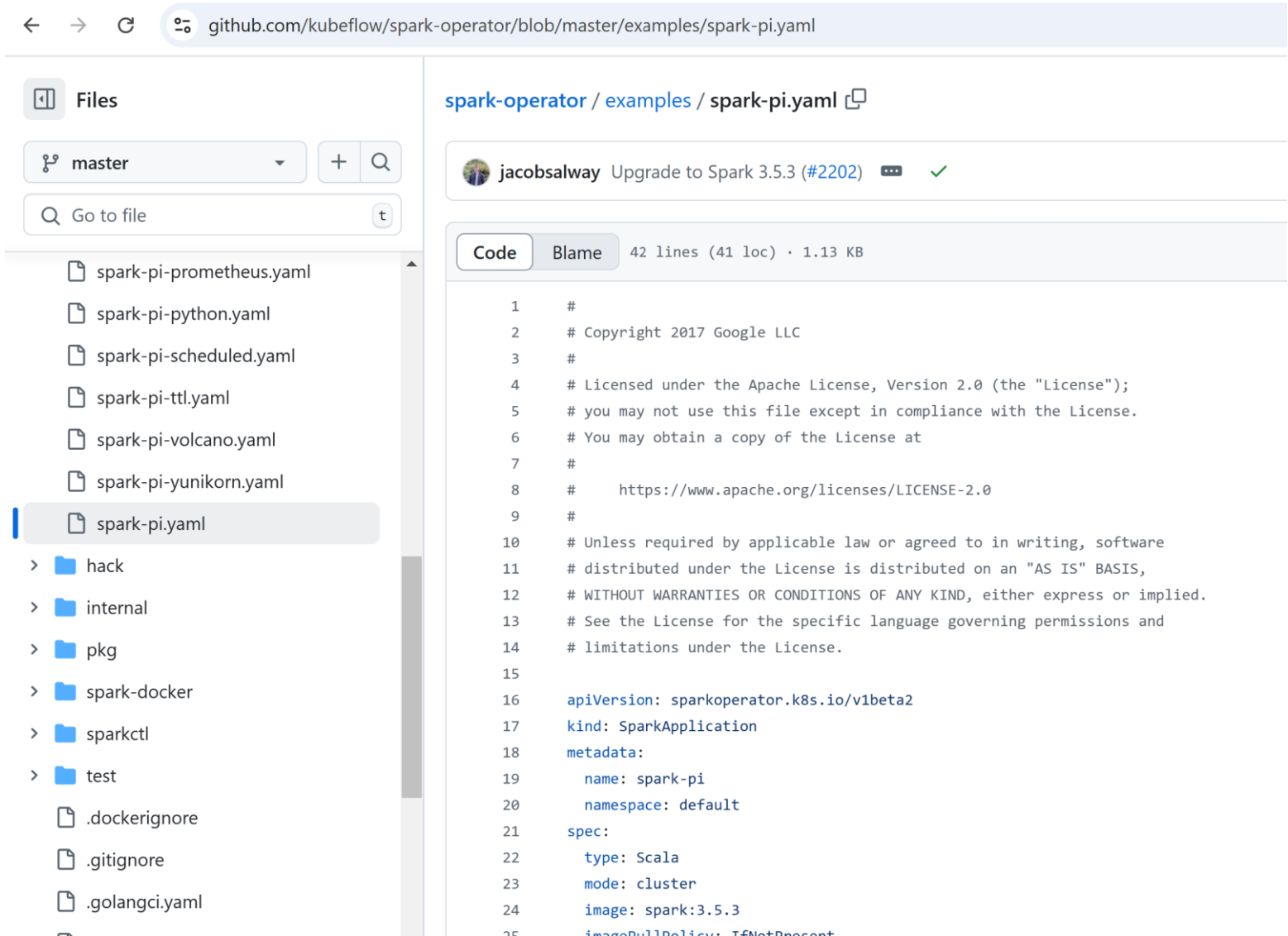
using "kubetctl apply -f <file.yaml>"

a.k.a. the K8s way
(instead of spark-submit)

https://www.kubeflow.org/docs/components/spark-operator/getting-started/#configuration



https://github.com/kubeflow/spark-operator/blob/master/examples/spark-pi.yaml



Download file + k apply

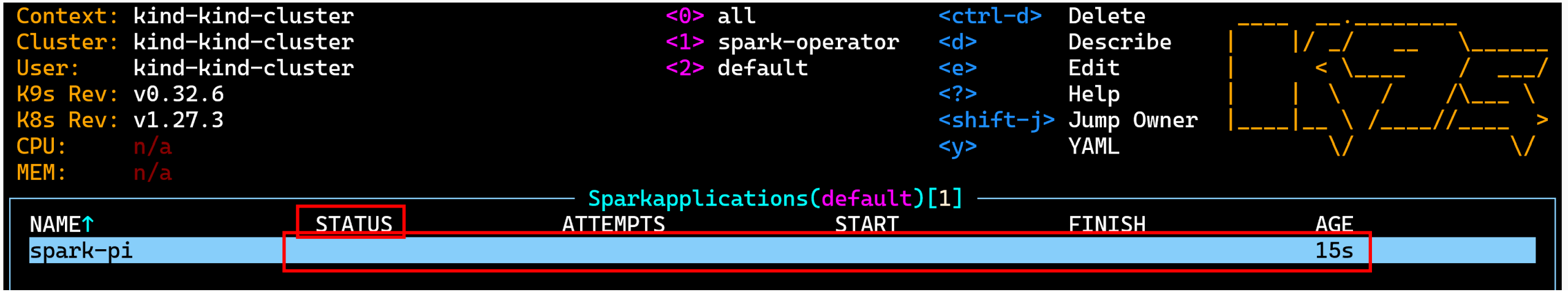
kubectl apply -f spark-pi.yaml

```
C:\apps\spark>kubectl apply -f  spark-pi.yaml
sparkapplication.sparkoperator.k8s.io/spark-pi created

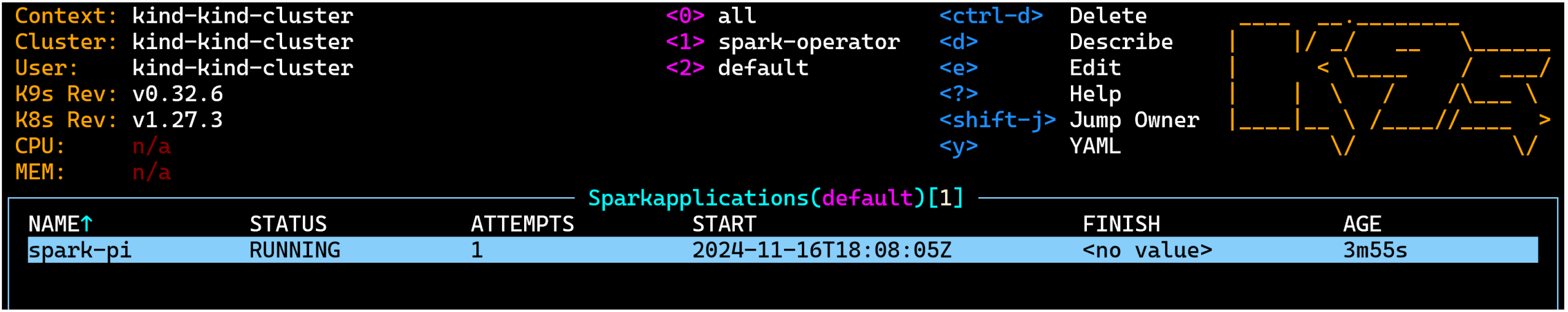
C:\apps\spark>
```

Step 1/3: sparkApplication status = " " (empty)

… spark-operator has not done spark-submit yet



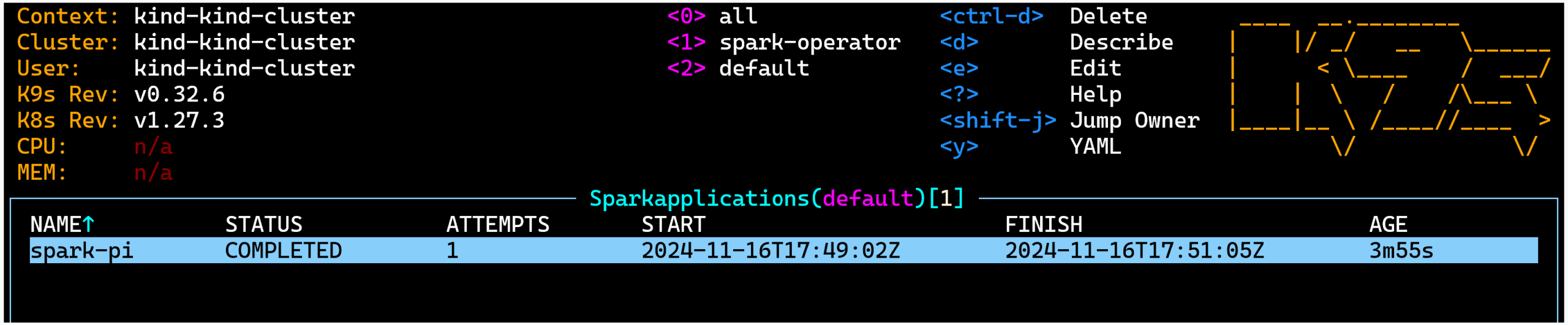
Step 2/3: sparkApplication status = RUNNING



kubectl get sparkApplication (list) status=RUNNING

```
C:\apps\spark>kubectl get sparkApplication
NAME       STATUS    ATTEMPTS   START                  FINISH       AGE
spark-pi   RUNNING   1          2024-11-16T18:08:05Z   <no value>   5m39s
```

Step 3/3: sparkApplication status=COMPLETED



kubectl get sparkApplication (list) … COMPLETED

```
C:\apps\spark>kubectl get sparkApplication
NAME       STATUS      ATTEMPTS   START                  FINISH                 AGE
spark-pi   COMPLETED   1          2024-11-16T17:26:00Z   2024-11-16T17:27:12Z   4m53s
```

kubectl apply AGAIN unchanged?
=> Does nothing

```
C:\apps\spark>kubectl apply -f  spark-pi.yaml
sparkapplication.sparkoperator.k8s.io/spark-pi unchanged
```

Unchanged => nothing is done.. K8s will not delete, nor restart anything
k apply does not "launch", it just write the desired state to be "a sparkApp should exist"

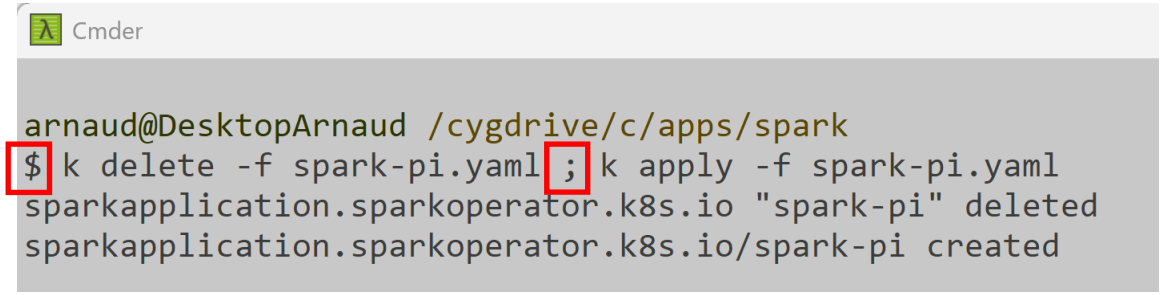
kubectl delete sparkApplication spark-pi

```
C:\apps\spark>kubectl delete sparkApplication spark-pi
sparkapplication.sparkoperator.k8s.io "spark-pi" deleted
```

Notice:
- you apply using file name "spark-pi.yaml"
- you can delete the sparkApp (only it) by its type + resource name "spark-pi"
- or by full name "sparkAplpication/spark-pi"
- or delete all resources from yaml file

```
C:\apps\spark>kubectl delete -f spark-pi.yaml
sparkapplication.sparkoperator.k8s.io "spark-pi" deleted
```

force relaunch on unix Bash
kubectl delete ; kubectl apply



NOTICE:
using unix "Bash $" shell, you can do "cmd1 ; cmd2" to execute sequentially on 1 line
But on cmd DOS c:>, you can't use ";" !
do not use "&&" which would stop on delete error !!
It is OK if the "kubectl delete " does not find anything to delete.

on Cmd Dos : && does not work you need "--ignore-no-found=true"

```
C:\apps\spark>kubectl delete -f spark-pi.yaml && kubectl apply -f  spark-pi.yaml
sparkapplication.sparkoperator.k8s.io "spark-pi" deleted
sparkapplication.sparkoperator.k8s.io/spark-pi created

C:\apps\spark>kubectl delete -f spark-pi.yaml
sparkapplication.sparkoperator.k8s.io "spark-pi" deleted

C:\apps\spark>kubectl delete -f spark-pi.yaml && kubectl apply -f  spark-pi.yaml
Error from server (NotFound): error when deleting "spark-pi.yaml": sparkapplications.sparkoperator.k8s.io "spark-pi" not found
```

OK when it existed: deleted , then created

ERROR when it did not exist: deleted, but not re-created !

```
C:\apps\spark>kubectl delete sparkApplication spark-pi --ignore-not-found=true && kubectl apply -f  spark-pi.yaml
sparkapplication.sparkoperator.k8s.io/spark-pi created

C:\apps\spark>kubectl delete sparkApplication spark-pi --ignore-not-found=true && kubectl apply -f  spark-pi.yaml
sparkapplication.sparkoperator.k8s.io "spark-pi" deleted
sparkapplication.sparkoperator.k8s.io/spark-pi created
```

OK

OK

Use helper ".cmd" File script

relaunch-spark-pi.cmd

```
1  @echo OFF
2
3  echo relaunching sparkApp
4
5  echo step 1/2: kubectl delete sparkApplication spark-pi
6  kubectl delete sparkApplication spark-pi --ignore-not-found=true
7
8  echo step 2/2: kubectl apply -f spark-pi.yaml
9  kubectl apply -f  spark-pi.yaml
```

```
C:\apps\spark>relaunch-spark-pi.cmd
relaunching sparkApp
step 1/2: kubectl delete sparkApplication spark-pi
sparkapplication.sparkoperator.k8s.io "spark-pi" deleted
step 2/2: kubectl apply -f spark-pi.yaml
sparkapplication.sparkoperator.k8s.io/spark-pi created
```

Next Steps

Try to change yaml file, to

- add "sparkConf" configuration
- change driver/executor config (memory, number of executors, etc..)
- add configMap + mount as volume for reading file
- add PVC
- add PVC for internal shuffling directories
- add toleration for NodePool taints
- etc etc

Try to open ingress equivalent of http://localhost:4040 (or port-forward)
Try to attach debugger (remote debug JVM process)

Next Advanced Steps (maybe Demo in Course)

create an Azure AKS cluster
launch from local "kubectl" command to remote Azure Kubernetes

configure several NodePools
  1 "on-demand" : small but "normal price" VM for the spark-driver
  1 "spot" : Big memory VMs but "Spot Instance" VM for executors

port-forward http://localhost:4040 for driver pod 4040

port-forward 5005 for remote debug from local PC in IntelliJ !!

mount volume from configMap
 + volume file from KeyVault (for password secrets)

use "managed-identity"
etc etc

Questions ?

arnaud.nauwynck@gmail.com